

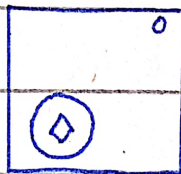
Means - End Analysis

- Means End Analysis (MEA) is a problem solving technique.
- Used to stimulate creativity.
- Means End Analysis is also a way of looking at the organisational planning and helps in achieving the end-goal.
- Such a mixed search algorithms allows us to solve the major part of a problem first and then go back and solve the small problem which arise while combining the major parts of the problem.
- Such a technique is called Means-End-Analysis.
- In the MEA process, first we find the difference the current state and goal state.
- Once we find the difference, then we apply an operator to reduce the difference.
- But sometimes it is possible that an

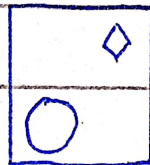
operator cannot be applied to the current state.

→ Hence we divide the current state into sub problems and then we apply an operator on sub problems such type of analysis is called operator sub goaling.

Example:-



Initial state



Goal State

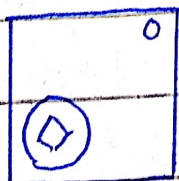
Operators for problem are:

→ Move

→ Delete

→ Expand

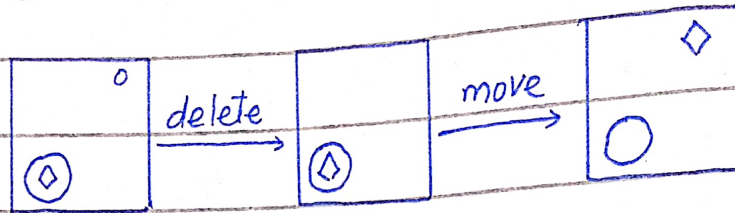
Evaluating initial state



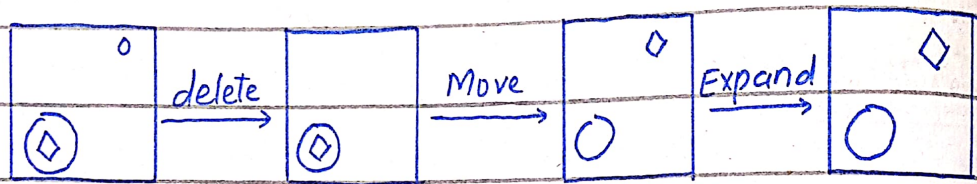
Applying delete operator



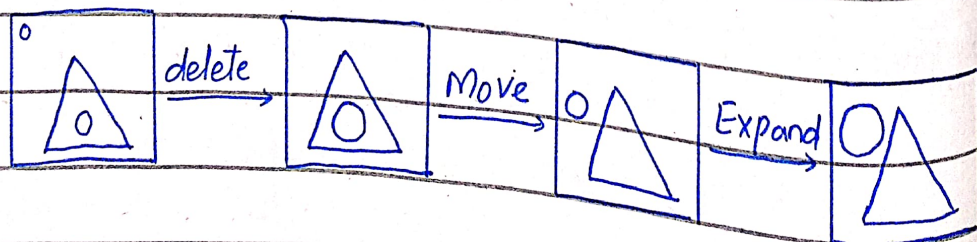
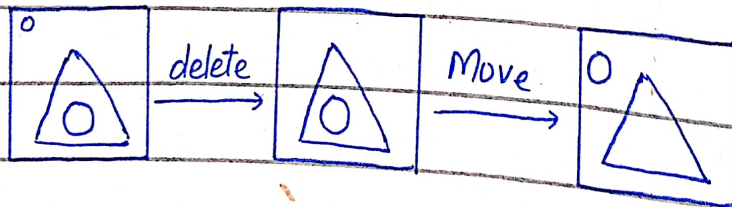
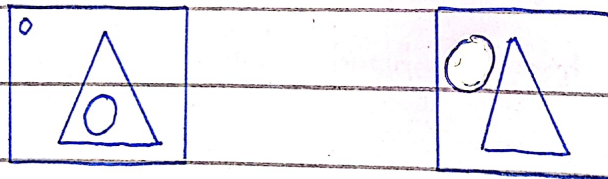
Applying move operator:-



Applying Expand operator:-



Example 2:-



Expert Systems:-

Expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.

Characteristics:-

- High Performance
- Understandable
- Reliable
- Highly Responsive

Capabilities:-

- Advising
- Instructing and assisting human in decision making
- Demonstrating
- Deriving a solution
- Diagnosing
- Explaining
- Interpreting input
- Predicting results
- Justify the conclusion
- Suggesting alternative options to a problem.

Incapabilities:-

- Substituting human decision makers
- Possessing human capabilities
- Producing accurate output for inadequate knowledge base
- Refining their own knowledge

Components of Expert System:-

- Knowledge Base
- Inference Engine
- User Interface